Fiscal Year:	FY 2016	**Task Last Updated:**	FY 12/06/2016
PI Name:	Monshausen, Gabriele Ph.D.		
Project Title:	Defining the Role of the Receptor Like Kinase FERONIA in Plant Architecture Development Under Mechanical Load		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802-5301	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/02/2013	**End Date:**	11/01/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date change to 11/1/2017 per NSSC information (Ed., 4/5/16) NOTE: Period of performance corrected to actual dates per NSSC information (Ed., 9/10/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	The shaping of plant architecture is a process that unfolds over the entire lifespan of a plant and is continually informed by developmental and environmental cues. On Earth, the most pervasive and constant of these environmental cues is gravity. Gravity is an important factor in guiding how plant organs are organized in space. However, gravity does not only provide directional information during plant development. As growing plants gain size and mass, the increasing weight of vertical stems and laterally projecting branches and leaves results in progressive mechanical self-loading. These mechanical loads can lead to buckling or breaking of plant organs unless they are structurally reinforced. The principal objective of this research proposal is to understand how mechanical loads imposed by gravity are translated into developmental programs that maintain structural integrity of the plant body. Specifically, we will investigate the role of the plasma membrane receptor-like kinase FERONIA in this process, as we have recently discovered that FERONIA is crucial to normal mechanical signaling and mechanical development of the model plant Arabidopsis thaliana (Mouse Ear Cress). We will employ fluorescence microscopy, growth kinematics, molecular biology, and genomic approaches to investigate the impact of variable gravity conditions on plant architecture development and to define how FERONIA interacts with mechanical signal transduction pathways. Results from such analyses are responsive to the goals of the NASA Research Announcement in addressing the fundamental question of how plants sense and react to gravity as a mechanical force to regulate growth and development. Understanding how FERONIA serves to help plants adapt under conditions of non-Earth-normal gravity may eventually provide a means for modulating mechanical strength in plants raised in microgravity conditions in the absence of mechanical self-load.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research addresses the question of how plants sense mechanical forces and how they utilize this information to modulate growth and tissue strength. Understanding how mechanical signaling, and specifically mechanical signaling through the receptor-like kinase FERONIA, affects the growth and development of plants will not only provide insight into how the reduced gravity environment of spaceflight will affect plant shoot architecture, but will also have translational potential for improving plant agriculture on Earth. Understanding the mechanisms that enhance plant tissue strength may eventually yield molecular targets for the manipulation of traits that allow plants to withstand wind and to penetrate compacted soils.
Task Progress:	The question of how the phytohormone auxin regulates plant cell expansion in an organ-specific manner has fascinated biologists since Darwin first predicted the existence of an ‘influence’ that controls plant directional movements. However, despite a wealth of research, the molecular identification of proteins involved in the earliest stages of auxin signal transduction has remained a challenge. We have identified the cyclic nucleotide-gated channel CNGC14 as a key regulator of auxin-induced Ca2+ signaling and control element of auxin-regulated cell expansion in roots. Cngc14 mutants exhibit a complete loss of rapid auxin-induced ion signaling and a clear delay in the earliest phase of auxin-induced root growth inhibition, both in response to exogenous auxin treatment and in response to endogenous auxin signaling during root gravitropism. These results provide a molecular mechanism linking auxin to short-term regulation of plant growth responses in the induction of Ca2+-dependent signal transduction. We propose that an ion signaling module consisting of Ca2+ and extracellular alkalinization is evolutionarily conserved and has been recruited by diverse plant stress signaling cascades to accelerate growth responses. Cellular perception of mechanical stress is of fundamental importance to living systems. In plants, environmental mechanical stresses shape plant architecture from the cellular level to the level of organ formation. We have previously identified a receptor-like kinase, FERONIA, as a key regulator of mechanical Ca2+ signaling and control element of root growth responses to environmental mechanical stresses and turgor-driven cellular expansion. We have now further found that FER is also an important regulator of seedling shoot responses to mechanical perturbation. Whether FER functions as a true mechanosensor remains to be established, but current evidence suggests that FER is directly involved in the early phases of mechanical signal transduction. We have developed computer-vision-assisted assays to monitor how mechanical signaling affects the mechanical properties of cell walls in living, intact organs at subcellular spatial and second temporal resolution.
Bibliography Type:	Description: (Last Updated: 12/06/2016)
Articles in Peer-reviewed Journals	Shih HW, DePew CL, Miller ND, Monshausen GB. "The cyclic nucleotide-gated channel CNGC14 regulates root gravitropism in Arabidopsis thaliana." Curr Biol. 2015 Dec 7;25(23):3119-25. http://dx.doi.org/10.1016/j.cub.2015.10.025 ; PubMed PMID: 26752079 , Dec-2015